

Helix OTA — OpenDesign Adoption (§11.4.162) + Host- Render Visual-Proof (§11.4.170) — Ready-to-Execute Plan

Revision: 1 **Last modified:** 2026-07-09T00:00:00Z **Trigger:** Follow-up to FINDINGS.md — produce a plan so work starts the moment the operator picks a canonical frontend. **Scope of this doc:** research/plan ONLY. No frontend code was modified (read-only on `clients/ota-manager/` and `dashboard/` per §11.4.122). The canonical-frontend choice is an OPERATOR decision (§11.4.66). **Anti-bluff (§11.4.6):** every stack claim below is cited from a file read this session. Every external claim carries a source URL + date in the Sources footer.

0. TL;DR

- **Recommendation:** make `clients/ota-manager/` the canonical frontend and the OpenDesign-refinement target. It already has a real design-token architecture (CSS-variable tokens + Tailwind theme mapping + a working light/dark toggle) — the exact seam OpenDesign tokens plug into. `dashboard/` uses inline-style primitives with no tokens and no dark mode, so adopting OpenDesign there is a UI rebuild, not a refinement.
- **OpenDesign canonical source: UNCONFIRMED.** §11.4.162 names “OpenDesign” but no repo/URL, and no OpenDesign submodule exists in `.gitmodules`. Closest evidenced candidate: “Open Design” (`open-design.ai` / `opendesigner.io`, GitHub `nexu-io/open-design`) — an agent-oriented design-system tool that emits **CSS-custom-property design tokens** (light/dark, palette/typography/spacing/component-level) — which matches the §11.4.162 description and maps cleanly onto `ota-manager`’s existing token surface. Operator must confirm the intended OpenDesign before install.
- **§11.4.170 host-render harness:** Storybook 8 (Vite builder) + `@storybook/addon-themes` (renders each story in light+dark) + **native**

Playwright toHaveScreenshot() for golden image-diff + a Tesseract-OCR/bounding-box **layout oracle** (no-overlap/clipping/off-screen), self-validated with golden-good/golden-bad fixtures per §11.4.107(10).

1. Confirmed stack of each frontend (read-only file evidence)

clients/ota-manager/ — “OTA Manager” SPA + Tauri desktop

Evidence: package.json, tailwind.config.ts, components.json, src/index.css, src-tauri/tauri.conf.json, src/stores/ui-store.ts, src/features/layout/topbar.tsx, vitest.config.ts, src/ tree.

Aspect	Evidence
Framework	React 19.0 + React-DOM 19, TypeScript 5.7, Vite 6 , pnpm (pnpm-lock.yaml, pnpm-workspace.yaml)
Component primitives	shadcn/ui (components.json \$schema=shadcn.com, style default, baseColor slate, cssVariables:true) over Radix (@radix-ui/react-{dialog,dropdown-menu,progress,tabs}) + class-variance-authority + clsx + tailwind-merge + lucide-react. 15 primitives under src/components/ui/ (alert-dialog, badge, button, card, dialog, dropdown-menu, form, input, progress, select, sheet, skeleton, table, tabs, toast) + data-table/.
Styling / tokens	Tailwind CSS 4 (@tailwindcss/vite, @tailwindcss/postcss, tailwindcss@^4). tailwind.config.ts darkMode: "class", theme colors mapped to hsl(var(--token)).

Aspect	Evidence
Theme switching	src/index.css defines a full HSL CSS-variable token set for :root (dark default) and .light (background/foreground/card/popover/primary/secondary/muted/accent/destructive/border/input/ring/radius + sidebar-*). This IS a real light+dark token system (not OpenDesign's). Working: ui-store.ts Zustand store theme: "dark" + toggleTheme (persisted); topbar.tsx renders a "Toggle theme" Button wired to toggleTheme.
Data / routing / forms	TanStack Query 5 + TanStack Router 1 (+ generated route-tree.gen.ts) + TanStack Table 8; react-router-dom@7 also present; Zustand 5; react-hook-form 7 + @hookform/resolvers + zod 3; axios.
Feature modules	8: auth, layout, dashboard, devices, releases, deployments, groups, audit (src/features/*).
Desktop/mobile reach	Tauri v2 desktop (src-auri/, tauri.conf.json productName "Helix OTA Manager", bundle.targets:"all", single 1280×800 window). @auri-apps/plugin-{fs,shell}. Mobile is NOT scaffolded — src-auri/gen/schemas/ has only desktop-schema.json + macOS-schema.json; no gen/android OR gen/apple. FINDINGS' "experimental Android/iOS shell" overstates: config aspires (targets:"all"), platforms are not generated.
Embed path	Built dist/ is embedded into the Go server as manager-dist/ and served at /manager (server/

Aspect	Evidence
Test harness	internal/api/embed.go, per FINDINGS) — a real production delivery path. Vitest 4 + RTL only (vitest.config.ts, src/test/, src/__tests__/). No e2e, no Playwright, no visual/snapshot harness.

dashboard/ — “Operator Dashboard” MVP

Evidence: package.json, src/components/ui.tsx, src/tree, e2e/tree, playwright.config.ts, BROWSER_TEST_EVIDENCE.md.

Aspect	Evidence
Framework	React 18.3 + React-DOM 18.3, TypeScript 5.6, Vite 5, react-router-dom 6, npm (package-lock.json). Version 1.0.0-mvp. Hand-rolled src/components/ui.tsx — self-contained primitives (Card/Button/Field/TextInput/...) styled with inline style={CSSProperties}. Header comment: names an
Component primitives	“UNVERIFIED UI-Components - React catalogue brick... NOT a confirmed dependency”; primitives carry NO API knowledge (decoupled). No Tailwind, no CSS variables, no design tokens, no dark mode.
Styling / tokens	None — inline JS style objects. No theming system.
Screens	8: Login, ArtifactUpload, Releases, Deployments, Fleet, Overview, Groups, Audit (src/screens/*).
Test harness	Strong. Vitest 2 unit tests per screen + Playwright 1.60 e2e (e2e/: smoke, ally.spec.ts with

Aspect	Evidence
Named as canonical?	@axe-core/playwright, fleet-detail, deployments, groups, populated-detail, releases, record-all-screens.spec.ts) + vitest-axe + captured BROWSER_TEST_EVIDENCE.md (live-app screenshots/video/axe). Satisfies §11.4.153/.158/.159 live-device recording; does NOT satisfy §11.4.170 (host-side, no-running-app, component-level). Yes in prose — root README + formal design doc docs/research/main_specs/1.0.0-mvp/dashboard/dashboard_design.md.

Decision matrix — as the OpenDesign-refinement target

Criterion	clients/ota-manager	dashboard
Component maturity	shadcn/ui + Radix (15 primitives, a11y-grade)	hand-rolled inline-style primitives
Design-token readiness (the seam OpenDesign plugs into)	Already token-driven (CSS vars + Tailwind theme + light/dark toggle)	None — inline styles; would need full rewrite
Light/dark themes	Working (Zustand toggle + .light/:root)	Absent
React / build recency	React 19 + Vite 6 (newer)	React 18 + Vite 5
Desktop reach	Tauri v2 desktop (native)	browser SPA only
Mobile reach	aspirational (targets:"all"), not scaffolded	none
Production embed	Served at /manager in Go server	standalone dev server
Test coverage	Weak — Vitest/RTL only, no e2e/visual	

Criterion	<code>clients/ota-manager</code>	dashboard
		Strong — Playwright + axe + captured evidence
Recency (last commit)	2026-06-20	2026-06-20 (tie)
OpenDesign adoption cost	Refinement (map tokens onto existing CSS-var surface)	Rebuild (introduce a token system where none exists)

Recommendation (evidenced, §11.4.6): adopt `clients/ota-manager` as canonical + OpenDesign target. Rationale: (1) OpenDesign is a design-**token** system; ota-manager already has the token architecture OpenDesign feeds, so adoption is a token-substitution/refinement, whereas dashboard has no tokens at all (a rebuild). (2) It is the more capable, more modern product surface (shadcn/Radix, React 19, native desktop, real light+dark) and is the one already wired into the production server at `/manager`. (3) ota-manager’s only real deficit — test coverage — is *exactly* what the §11.4.170 harness in this plan supplies, and dashboard’s strong Playwright/axe patterns are **portable** (lift its e2e helpers into ota-manager). The reverse gap (dashboard’s missing UI/token maturity) is not portable. Recency is a tie, so it does not decide.

Honest boundary: this is a technical recommendation, not the decision. dashboard is named in the README + a formal design doc, so retiring it is a §11.4.122 no-silent-removal event requiring explicit operator approval + an Obsolete (→ Fixed.md) item (§11.4.90). The operator may also choose to keep both.

A. Canonical-frontend recommendation + operator option-set (AskUserQuestion)

Put this question to the operator via AskUserQuestion (§11.4.66) **before** any OpenDesign install. Recommended option **bolded**.

Question: “Two React frontends exist (`clients/ota-manager` = shadcn/ui + Tailwind 4 + Tauri desktop, token-ready, embedded at `/manager`; dashboard = MVP with inline-style primitives but strong Playwright/axe tests). Which is canonical for OpenDesign (§11.4.162) adoption + the §11.4.170 visual-proof harness?”

#	Option	What it means	Consequence
1	ota-manager canonical; retire dashboard (RECOMMENDED)	Adopt OpenDesign + harness on ota-manager; port dashboard's Playwright/axe patterns over; mark dashboard Obsolete (→ Fixed.md) (§11.4.90) after this approval	One surface to maintain; token refinement not rebuild; loses nothing (test patterns ported)
2	ota-manager canonical; keep dashboard as-is	Adopt on ota-manager; leave dashboard maintained but un-refined (or backlog it)	Two surfaces; dashboard stays token-less until separately funded
3	dashboard canonical; retire ota-manager	Rebuild dashboard's UI onto a token system, then adopt OpenDesign; retire ota-manager (§11.4.122 approval)	Higher cost (UI rebuild); loses native desktop + /manager embed
4	Keep BOTH, refine BOTH	Adopt OpenDesign + harness on each (shared token package)	Highest cost; maximal reach

The three sub-steps below (B, C) are written for the RECOMMENDED target ota-manager; if the operator picks option 3/4 the same B/C templates apply to dashboard after its token-system rebuild.

B. OpenDesign adoption steps for clients/ ota-manager

B0. Resolve the OpenDesign source (BLOCKING — operator confirm)

§11.4.162 mandates “OpenDesign” but names no repo; there is **no OpenDesign entry in .gitmodules** and no vasic-digital/OpenDesign in the owned-org set. Research (Sources) found several distinct “OpenDesign” projects:

- **nexu-io/open-design (open-design.ai / opendesigner.io)** — CLOSEST MATCH. Agent-oriented design-system authoring tool; emits **CSS-custom-property design tokens** across a ~12-section schema (Surface, Text, Border, Accent, Semantic, Typography, Spacing, Radius, Elevation, Focus, Motion, Layout) with light/dark — aligns with §11.4.162’s “color palette (light+dark), typography, spacing, component-level tokens.” Installed by `git clone + npm tools-dev daemon (SQLite .od/, timestamped token/artifact renders)`; **not** a conventional `npm install` library.
- `opendesigndev/open-design-framework (@opendesign/react, “octopus” format)` — a design-**file** (Figma/Sketch) reader/renderer, **NOT** a token/theme system. Rejected.
- `manalkaff/opendesign (“claude.ai/design open-sourced”)`, `npm open-design-system`, `@opengovsg/design-system-react` — unrelated / not evidenced as the mandate’s referent.

Action: ask the operator (AskUserQuestion) to confirm WHICH OpenDesign §11.4.162 means. Present `nexu-io/open-design` as the recommended candidate; alternative = “it is meant to be an owned `vasic-digital/OpenDesign` submodule to be created per §11.4.28.” Do not install until confirmed (§11.4.6 no-guessing). Once confirmed, add it per §11.4.28 layout + §11.4.36 `install_upstreams`.

B1. Consume OpenDesign tokens (the model is source-agnostic)

Regardless of which OpenDesign is confirmed, its output is a **set of CSS custom properties per theme** — which is precisely `ota-manager`’s existing surface. Steps: 1. Generate/export OpenDesign’s token set for the **Helix brand** in light + dark (palette, typography, spacing, radius, elevation, focus, motion), sourced from canonical Helix brand assets. 2. Land the exported tokens as the single source of truth in `src/index.css`

(or a new `src/styles/opendesign.tokens.css` @import-ed first), replacing the current ad-hoc HSL values while keeping the SAME variable names the Tailwind theme already maps (`--background`, `--foreground`, `--primary`, ... in `tailwind.config.ts`) so no component churns. Add any NEW OpenDesign token families (elevation/focus/motion/spacing-scale) as new variables + Tailwind theme extensions. 3. Preserve the `:root(dark)/.light + darkMode: "class"` + Zustand-toggle mechanism; OpenDesign light/dark map onto these two class scopes. Verify the topbar toggle still flips both. 4. Re-skin the 15 shadcn primitives ONLY via token values (no per-component hard-coded colors); every primitive must ship a light AND dark variant (§11.4.162) and MUST NOT overlap/overlay labels (§11.4.162) — proven by section C, not by eye.

B2. §11.4.74 extend-upstream path for gaps

When ota-manager needs a pattern OpenDesign lacks (an OTA-specific token — e.g. deployment-state semantic colors, device-health status ramp — or a missing component token): 1. Do **not** fork project-locally and diverge silently. Follow §11.4.74/§11.4.28: extend the OpenDesign **upstream** (fork the confirmed repo, add the pattern, PR upstream) OR, if OpenDesign is an owned `vasic-digital/*` submodule, add the pattern there and bump the pointer (§11.4.26 step 7). 2. Keep the token package project-agnostic (§11.4.28 decoupling) — Helix-specific values are injected via the consumer's token file, never hard-coded into the shared OpenDesign package. 3. Every extension ships light+dark + is covered by the section-C harness before merge.

B3. Governance / tracking

- Correct the stale CLAUDE.md line “§11.4.162 latent until this project ships a UI surface” — the surfaces exist; §11.4.162 + §11.4.170 are ACTIVE (tracked adoption item, §11.4.93 workable-items DB).
 - Register the adoption as workable items (ATM-NNN) with §11.4.171 ≥5-7-sentence descriptions.
 - If option 1/3 chosen: file the retirement as Obsolete (→ `Fixed.md`) with operator-approval citation (§11.4.90/§11.4.122).
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C. §11.4.170 host-render visual-proof harness (shadcn / Vite / React 19 / Tailwind 4)

§11.4.170 requires: the real component rendered to a **PNG on the host** (no device/emulator/running app) for **every screen × state × {light, dark}**, dual-validated by (i) golden image-diff AND (ii) an OCR/vision oracle reading rendered text + labels + control bounds (no overlap / label-over-label / clipping / off-screen / collapsed-or-giant widget). Value/token-equality unit tests are FORBIDDEN as the sole proof.

C1. Tooling (named, cited)

Layer	Tool	Why
Component workshop / host render	Storybook 8 with the Vite builder (@storybook/react-vite)	Renders each shadcn primitive + each feature screen in isolation on the host; reuses ota-manager's Vite 6 config. Story = the “component × state” unit.
Light+dark rendering	@storybook/addon-themes with ThemeByClassName decorator (toggles .light/.dark — matches Tailwind darkMode: "class")	Produces the {light,dark} axis automatically: one story → two theme renders.
Golden image-diff (PNG per screen×state×theme)	Native Playwright toHaveScreenshot() driven against Storybook's iframe.html?id=<story>&globals=theme:<light\ dark> (via @storybook/test-runner, Playwright-based, OR a thin standalone Playwright project)	Built-in pixelmatch golden diff, local + free, deterministic; stores __screenshots__ goldens. Note: toHaveScreenshot() works only in native Playwright , NOT Vitest browser mode — so the visual layer is a Playwright project,

Layer	Tool	Why
OCR / vision layout oracle	Tesseract (system tesseract or tesseract.js) reading the captured PNG with per-word confidence + bounding boxes, PLUS Playwright boundingBox() + accessibility-tree bounds cross-check	not the Vitest addon. Satisfies §11.4.170(ii)/§11.4.107(12): asserts expected labels present at expected ROI (per-word confidence floor), and computes geometric checks: no bbox overlap between distinct controls, text bbox $\subseteq$ container bbox (no clipping), bounds on-screen and neither collapsed ($\approx 0\text{px}$) nor giant-unbounded.
Interaction/unit (supplementary, NOT the proof)	existing Vitest 4 + RTL ; optionally Storybook Vitest addon (Playwright provider) for play-function interaction	Complements per §11.4.170 — may supplement, never substitute the rendered-pixel proof.

Rejected/optional: Chromatic/Percy (cloud, paid, and §11.4.156 disables CI) — local Playwright screenshots are the default; a cloud service is optional and operator-gated. Roborazzi/Paparazzi are Android-Compose tools — N/A to this React stack.

C2. Matrix + procedure (per §11.4.170)

1. **Enumerate the matrix:** every shadcn primitive state (button: default/hover/focus/disabled/loading; input: empty/filled/error; dialog/sheet: open; table/data-table: empty/populated/loading; toast; badge variants; ...) + every feature screen (auth/dashboard/devices/releases/deployments/groups/audit) $\times$ {light, dark}. Each cell = one Storybook story rendered in one theme.

2. **Render → PNG (host):** Playwright loads the story iframe with the theme global, waits for fonts/animations to settle, `toHaveScreenshot()` → golden PNG.
3. **Golden image-diff:** subsequent runs diff against the committed golden; a diff over tolerance FAILs (the “giant broken button” regression that value-equality tests miss — the §11.4.170 forensic case — is caught here).
4. **OCR/vision oracle:** run Tesseract + bbox checks on the SAME PNG; assert label presence/ROI + no overlap/clipping/off-screen/collapsed/giant. FAIL pinpoints the offending element.
5. **Verdict:** a cell PASSes only if BOTH the image-diff AND the oracle pass; emit `ab_pass_with_evidence` citing the PNG + oracle report path (`video_display/subtitle_render-class` evidence per §11.4.69). Evidence under `docs/qa/<run-id>/opendesign_visual/`.

C3. Self-validated analyzer (§11.4.107(10)) — the oracle must not bluff

Ship, wired into meta-test: - **golden-good** fixture PNG (clean layout) → oracle MUST PASS. - **golden-bad** fixtures, one per failure family → oracle MUST FAIL, naming the family: (a) overlapping labels, (b) clipped text, (c) off-screen element, (d) collapsed (0px) widget, (e) giant-unbounded widget, (f) missing expected label. - **negative control:** a legitimately-different-but-valid layout → PASS (false-positive guard). An analyzer that PASSes any golden-bad, or FAILs golden-good/negative-control, is itself a §11.4 bluff (release blocker).

C4. Four-layer coverage (§11.4.4(b)) + gates

- **Pre-build gate** (CM-AV-LIVENESS-class, here CM-HOST-RENDERED-UI-VISUAL-PROOF): assert every UI component/screen has a story covering each state×{light,dark}, a committed golden, and an oracle assertion (not a value-equality test as sole proof).
- **Runtime/host test:** the Playwright+OCR run above.
- **Paired §1.1 mutation:** (i) break a component’s layout (overlap/clip) → visual-diff + oracle MUST FAIL; (ii) strip the oracle’s overlap check → self-validation golden-bad MUST FAIL.
- **HelixQA Challenge** entry referencing the run.
- **Determinism** (§11.4.50): pin fonts, disable animations, fixed viewport/DPR so N runs are byte-stable.

C5. Portability note

Lift dashboard/e2e/helpers.ts + a11y.spec.ts (@axe-core/playwright) patterns into ota-manager: run **axe** on each Storybook story render as a third oracle (accessibility) alongside image-diff + OCR — dashboard’s strongest asset, ported (supports option 1’s “lose nothing”).

D. Sources verified 2026-07-09

- Open Design (candidate) install/CLI + token model — <https://opendesigner.io/quickstart> (verified 2026-07-09)
- Open Design (candidate) schema/sections + CSS-custom-property tokens + light/dark — <https://open-design.ai/plugins/design-system-opencode-ai/> (verified 2026-07-09)
- Open Design GitHub repo (nexu-io/open-design) — <https://github.com/nexu-io/open-design> (verified 2026-07-09)
- OpenDesign disambiguation — octopus/design-file SDK (rejected) — <https://github.com/opendesigndev/open-design-framework> (verified 2026-07-09)
- OpenDesign disambiguation — claude.ai/design clone — <https://github.com/manalkaff/opendesign> (verified 2026-07-09)
- Playwright toHaveScreenshot() visual regression (built-in golden diff) — <https://bug0.com/knowledge-base/playwright-visual-regression-testing> (verified 2026-07-09)
- Storybook + Playwright visual regression (free, local) — <https://markus.oberlehner.net/blog/running-visual-regression-tests-with-storybook-and-playwright-for-free> (verified 2026-07-09)
- Storybook Vitest addon (Playwright provider) + toHaveScreenshot limitation (native Playwright only) — <https://storybook.js.org/docs/writing-tests/integrations/vitest-addon> (verified 2026-07-09)
- 2026 component-testing stack (Vitest + Storybook 8 + Playwright) — <https://www.pkgpulse.com/guides/playwright-component-vs-storybook-testing-2026> (verified 2026-07-09)
- Design-token light/dark modeling reference — <https://medium.com/design-bootcamp/color-tokens-guide-to-light-and-dark-modes-in-design-systems-146ab33023ac> (verified 2026-07-09)

Negative finding (§11.4.99): No official OpenDesign source names a vasic-digital/OpenDesign OR HelixDevelopment/OpenDesign repo, and none is in this project’s .gitmodules. The §11.4.162 mandate does not

cite a URL. Therefore the OpenDesign canonical source is **UNCONFIRMED**; nexu-io/open-design is the closest evidenced candidate and MUST be operator-confirmed (step B0) before install.